

Heart rate extraction based on eigenvalues using UWB impulse radar remote sensing

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ABSTRACT

Ultra wideband impulse (UWB) radar for vital signs measurement applications is getting much attention because it is a contactless method. However, the heartbeat signal is usually be perturbed in strong noisy signals from stationary and non-stationary objects. In addition, the respiration signal, which has a much higher intensity than the heartbeat signal, makes it challenging to extract the heartbeat signal. Thus, this paper proposes an extraction method of heartbeat signal and frequency using eigenvalues. Eigenvalues of the vital signal were found to be strongly correlated with the fundamental and harmonic frequencies of respiration and heartbeat. Experimental results show that the proposed method could improve the signal-to-noise ratio of over 18 dB for different distances from 0.5 m to 3.0 m and different orientations of the person. Furthermore, the original signal of respiration and heartbeat can be reconstructed by the selection of the eigenvalues and the eigenvectors.

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1. Introduction

Ultra-wideband impulse (UWB) radars are using in many indoor applications of smart home and a smart city such as target localization, human detection, fall detection, human gait analysis, and vital sign detection [1–6]. UWB radars have advantages of low power consumption, high spatial resolution, fewer interferences, and remote sensing. The applications of UWB radars for vital signs detection are getting much attention nowadays. UWB radar can be applied for life detection in hazard environments such as in ruins of house or ruins of mines because of the ability of penetration through objects [7]. UWB radar also can be used in healthcare application for remote monitoring of respiration and heart rate of a patient whom stationary stay on a bed [8–15]. It is also possible to apply for the non-stationary cases while a person is doing slow activities such as speaking, moving head, moving hand, or disk washing [16]. Another interesting UWB radar application in localization and detection the number of people sitting in a room was also developed in [17]; respiration signal pattern was used as a key for the localization and detection number of people.

UWB radar measures the vital signs by measuring the small movement of the thorax, which caused by heartbeat and respira-

tory activities. Vital signs signal of UWB radar is usually perturbed by noise from both stationary and non-stationary objects in the environment. Thus, UWB radar should accompany with a high-performance signal processing algorithm to accurately extract the respiration and heart rate. Singular value decomposition (SVD) was used to remove the stationary signal of a wall and other static objects [8,18]. A higher-order cumulant (HOC) model was used with a fast Fourier transform (FFT) to reduce the harmonic frequencies of respiration to extract heartbeat frequency [19]. Continuous wavelet transforms (CWTs) were used to extract heartbeat signal in the mixing signal with respiration and noise signal [20]. Ensemble decomposition modes (EMDs) were used to precisely extract respiration and heartbeat signal by decomposing the signal into its intrinsic components [21].

In this paper, a frequency extraction method will be proposed based on eigenvalues and its orders for the extraction of respiration and heart rate. The vital signal which is mixed by respiration signal, heartbeat signal, and noise signal is firstly constructed to a Hankel's matrix. Then the eigenvalues of the correlation matrix of the Hankel's matrix will be taken into account for accurate extraction of the vital signal frequencies including heart rate. The peaks in the frequency domain were found to have a good correlation with the eigenvalues and their orders. The performance of the proposed method will be compared with the direct radio frequency (RF) sampling method and a reference photoplethysmography (PPG) contact measurement sensor.

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2. Principle vital sign measurement using UWB radar

2.1. UWB radar signal model

This section summaries the result that was presented in the references [13,22]. Support, the nominal distance from the person to the UWB radar receiver, is d_0 . It is assumed that the motions of thorax caused by respiration activity and heartbeat activity are both sinusoids with amplitudes d_r and d_h , and frequencies f_r and f_h , respectively. Then, the distance from the thorax to the UWB radar is the summation $d(t)$, as expressed in Eq. (1); where t is the slow cross-range time.

$$d(t) = d_0 + d_r \sin(2\pi f_r t) + d_h \sin(2\pi f_h t) \quad (1)$$

The signal received at the receiver of radar, which is the summation of the reflected signal at the person and multipath components can be modeled as Eq. (2) [13,22].

$$x(t, \tau) = \sum_i A_i(p(\tau - \tau_i) + A_d(p(\tau - \tau_d(t))) \quad (2)$$

where $p(t)$ is the collected pulse with a carrier frequency ν_c ; A_i and τ_i denote the strength and delay time of multipath components, and A_d and τ_d denote the amplitude of the collected pulse and time-of-arrival (ToA) from the person, respectively [13,22]. τ_d is the ToA of two ways, which can be calculated by the relationship with distance $d(t)$ and speed of light c as Eq. (3).

$$\tau_d(t) = 2d(t)/c = \tau_0 + \tau_r \sin(2\pi f_r t) + \tau_h \sin(2\pi f_h t) \quad (3)$$

where τ_r and τ_h are time delays amplitude of respiratory and cardiac activities, respectively; τ_0 is nominal time delay at the person. It is noted that the static objects, transmitter and receiver antennas interference and stationary parts of person body can produce multipath components, which are DC components in the signal $x_d(t, \tau)$. Finally, the signal of respiratory and cardiac activities at the receiver without DC component can be expressed at Eq. (4), which is a 2-dimensional matrix signal.

$$x_d(t, \tau) = A_d(p(\tau - \tau_d(t))) \quad (4)$$

2.2. Direct RF sampling method

The direct RF sampling method was developed in [13,22] and will be extended with the application of chirp z-transform (CZT). With a high sampling frequency over the Nyquist's frequency, the received pulses could be directly sampled with a simple hardware configuration. The received signal will be formed in a 2D matrix with analog form in Eq. (4). To find respiration and heartbeat frequency, the signal $x_d(t, \tau)$ could be transformed into the frequency domain in the slow-time direction by Fourier transform $F(f, \tau)$, as expressed in Eq. (5).

$$F(f, \tau) = \int_{-\infty}^{+\infty} x_d(t, \tau) e^{-j2\pi f t} dt \quad (5)$$

Through the forward and backward transformation, then $F(f, \tau)$ could be represented in a form function of respiration and heartbeat frequency, as expressed in Eq. (6) [13,22]; where, ν and f are Fourier transform components in the fast and slow- time, respectively. $P(\nu)$ is the Fourier transform of pulse $p(t, \tau)$ in the fast-time direction τ ; $J_k(\beta_r \nu)$ and $J_l(\beta_h \nu)$ are Bessel function with $\beta_r = 2\pi \tau_r$ and $\beta_h = 2\pi \tau_h$.

$$F(f, \tau) = A_d \sum_{-\infty}^{+\infty} \sum_{-\infty}^{+\infty} C_{kl}(\tau) \delta(f - kf_r - lf_h), \quad (6)$$

$$C_{kl}(\tau) = \int_{-\infty}^{+\infty} P(\nu) J_k(\beta_r \nu) J_l(\beta_h \nu) e^{-j2\pi \nu (\tau - \tau_0)} d\nu \quad (7)$$

with coefficients, C_{kl} is a function of the fast-time τ , as expressed in Eq. (7). And, the maximum energy of the $F(f, \tau)$ could be found at $\tau = \tau_0$ [13,22]. It is seen that the Fourier transform of the vital signal is a function of respiration frequency f_r , heartbeat frequency f_h , and harmonics respiration kf_r ($k > 1$), harmonic heartbeat lf_h ($l > 1$), and intermodulation frequency $kf_r + lf_h$. The mix of respiration and heartbeat signal and their harmonic components could make challenging to detect the heartbeat frequency.

After sampled and transferred to a PC, the vital signal could be analyzed in the frequency domain by applying discrete Fourier transform (DFT). However, the DFT analyzes the signal with a frequency resolution of f_s/N , where f_s is the sampling rate and N is the length of the vital signal. This frequency resolution could be low if the sampling rate is low and the signal length is short. To improve the frequency resolution in the frequency domain, CZT could be used instead of the DFT. The CZT is a zoomed version of DFT in a predefined range of frequency, which provides better frequency resolution than the DFT method. Support, the interested range of frequency is from f_{min} to f_{max} ($f_{min} < f_{max} < f_s$), then the frequency resolutions of CZT is $(f_{max} - f_{min})/N$. For instance, if f_s is 44 frames/s, N is 512 samples (frames), then the DFT resolution is 0.0859 Hz which respects to 5.1 BPM which is very large error compare to the normal respiration rate of 6–42 BPM; while the CZT resolution will be 0.0057 Hz (0.4 BPM), which is a quite a good resolution.

2.3. Eigenvalue based method

This section proposed eigenvalue based method for vital components extraction. The method is based on the principle component analysis method, which was developed for analysis of the vibration of mechanical system [23]. It is extent to apply for the vital signal in the extraction of respiration and heartbeat signals. The purpose is using eigenvalues to decompose the vital signal into different subspace signals of fundamental and harmonics of the respiration signal and heartbeat signal. It is also finding the subspace of noise for noise removal. Support, $x = [x_1, x_2, \dots, x_L]$ is a vector of the vital digitized signal of Eq. (4) in the slow-time direction at a fixed fast-time $\tau = \tau_0$, where L is a number of samples in the slow-time. A trajectory matrix $\mathbf{H}$ is formed as Eq. (8), which is also called Hankel's matrix. Matrix $\mathbf{H}$ has a size of $m \times n$ with $n = L - m + 1$, and m is chosen such that $2 \leq m \leq L$. In here, m is $L/2$ when L is even and $(L+1)/2$ when L is odd, which are known as an optimized value [23,24].

$$\mathbf{H} = \begin{bmatrix} x_1 & x_2 & \dots & x_n \\ x_2 & x_3 & \dots & x_{n+1} \\ \vdots & \vdots & \ddots & \vdots \\ x_m & x_{m+1} & \dots & x_L \end{bmatrix} \quad (8)$$

Covariance matrix ($\mathbf{C}$) of $\mathbf{H}$ is then computed, as expressed in Eq. (9). Eigenvalues λ_i and eigenvectors of covariance matrix $\mathbf{C}$ can be computed by SVD, as expressed in Eq. (10); where $\mathbf{U}$ and $\mathbf{V}$ are left- and right- unitary matrices, respectively. The columns of $\mathbf{U}$ and $\mathbf{V}$ are called left and right eigenvectors ($U_i = \mathbf{U}(:,i)$, $V_i = \mathbf{V}(:,i)$); and $\mathbf{S}$ contains eigenvalues in its diagonal, $\lambda = \text{diag}(\mathbf{S})$. Eigenvalues λ_i is then arranged in descending order, $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_m \geq 0$ with the higher value represents for, the higher intensity signal of matrix signal $\mathbf{H}$ [23,24].

$$\mathbf{C} = \frac{1}{n} \mathbf{H} \mathbf{H}^T \quad (9)$$

$$\mathbf{C} = \mathbf{U} \mathbf{S} \mathbf{V}^T \quad (10)$$

Trajectory matrix $\mathbf{H}$ can be decomposed into subcomponents as $\mathbf{H} = \mathbf{H}_1 + \mathbf{H}_2 + \dots + \mathbf{H}_m$, where $\mathbf{H}_i = \sqrt{\lambda_i} U_i V_i^T$ are called rank-one matrices. According to the properties of the rank-one

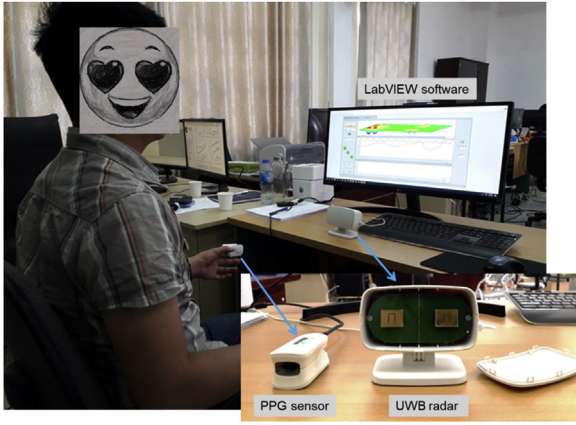


Fig. 1. Experimental setup.

Table 1
Specifications of the UWB radar.

Parameters	Specifications
Center frequency	4.6 GHz
Bandwidth	550 MHz
Distance resolution	2.03 cm
Supply voltage	5V (USB port)
Antennas	Directional type
Antennas beam	X-Z Plane: 75° Y-Z Plane: 80°
Max current consumption	350 mA
Output power	<−41.3 dBm
Frame rate	44.9 frames/s

matrices, matrix $\mathbf{H}$ can be grouped in a different form as $\mathbf{H} = \mathbf{H}_{g1} + \mathbf{H}_{g2} + \dots + \mathbf{H}_{gk}$. The rank-one matrices could be analyzed by the distribution of the eigenvalues λ_i . For instance, the first

two eigenvalues represent for respiration signal, the next two eigenvalues represent for heartbeat signal, and the remaining ones represent for the noise signal. Then, the trajectory matrix $\mathbf{H}$ is decomposed into three groups as $\mathbf{H} = \mathbf{H}_{resp} + \mathbf{H}_{heart} + \mathbf{H}_{noise}$,

$$\text{with } \mathbf{H}_{resp} = \sum_{i=1}^2 \sqrt{\lambda_i} \mathbf{U}_i \mathbf{V}_i^T, \mathbf{H}_{heart} = \sum_{i=3}^4 \sqrt{\lambda_i} \mathbf{U}_i \mathbf{V}_i^T, \text{ and } \mathbf{H}_{noise} = \sum_{i=5}^m \sqrt{\lambda_i} \mathbf{U}_i \mathbf{V}_i^T.$$

After decomposing the trajectory matrix $\mathbf{H}$ into the subcomponents, the time-series signal of the subcomponents could be reconstructed by the diagonal averaging. The reconstructed signal of the k -th group $\mathbf{H}_{gk}$ can be found by Eq. (11) [23]. Thus, by the selection of the eigenvalues (λ_i) and eigenvectors ($\mathbf{U}_i, \mathbf{V}_i$), the respiration and heartbeat signal could be accurately extracted from the noise signal. The collections ($\sqrt{\lambda_i}, \mathbf{U}_i, \mathbf{V}_i$) are called eigentriples.

$$\hat{x}^{(k)}(l) = \begin{cases} \frac{1}{l} \sum_{i=1}^l H_{gk}(i, l-i+1) & \text{for } 1 \leq l < m \\ \sum_{i=1}^m H_{gk}(i, l-i+1) \frac{1}{m} & \text{for } m \leq l < n \\ \frac{1}{L-j+1} \sum_{i=l-n+1}^m H_{gk}(i, l-i+1) & \text{for } n \leq l \leq L \end{cases} \quad (11)$$

3. Experimental analysis & discussion

3.1. Experimental setup

To demonstrate the proposed methods, UWB radar was used in the experiment to measure a healthy person, as shown in Fig. 1.

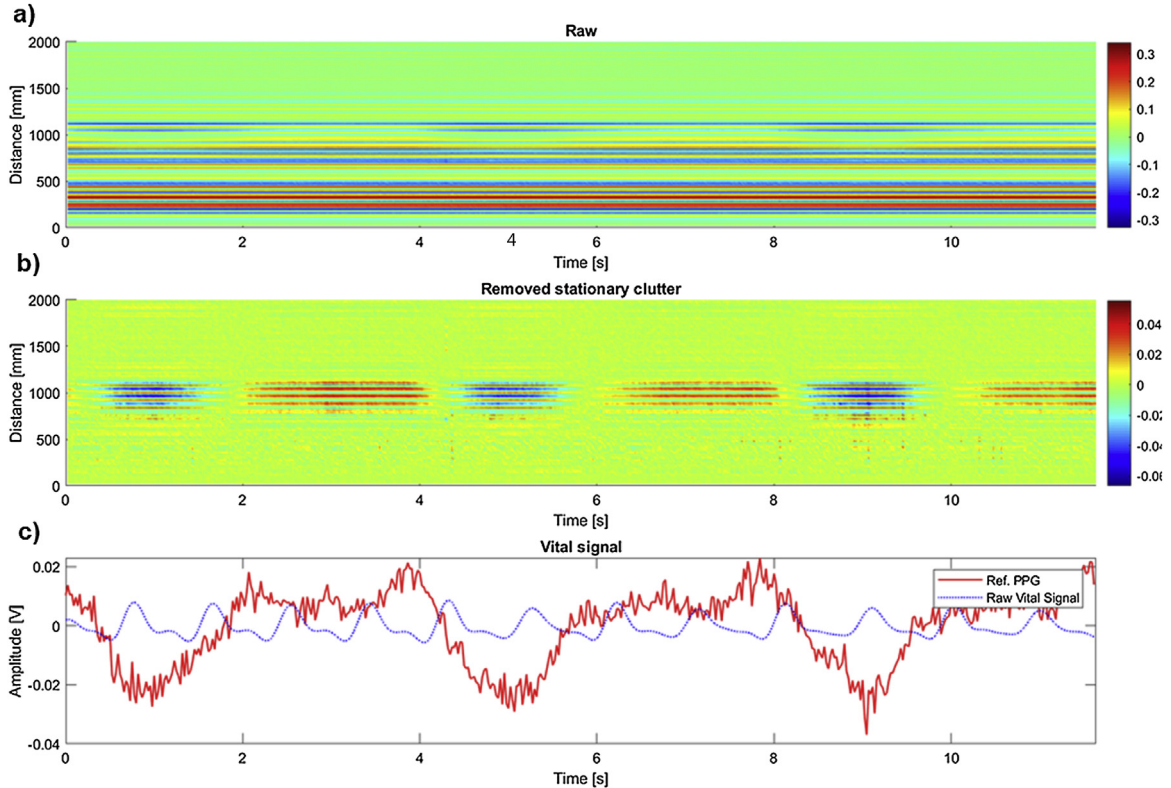


Fig. 2. Vital signal of a person sitting at 1.0 m: a) raw signal, b) removed stationary clutter, and c) vital signal waveform with reference heartbeat signal of the PPG sensor.

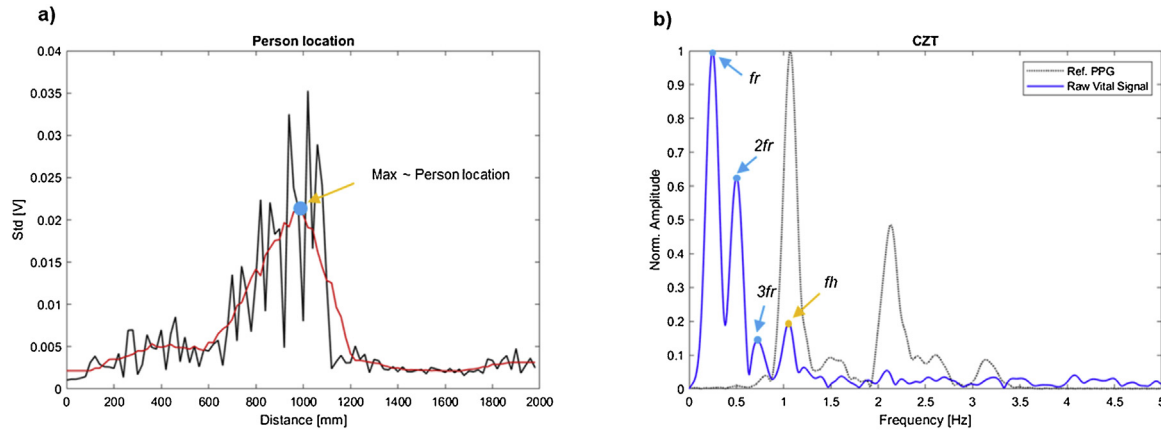


Fig. 3. Finding a person location by the standard deviation (a), and CZT of the vital signal by UWB radar at the person position and reference heartbeat signal of PPG sensor (b).

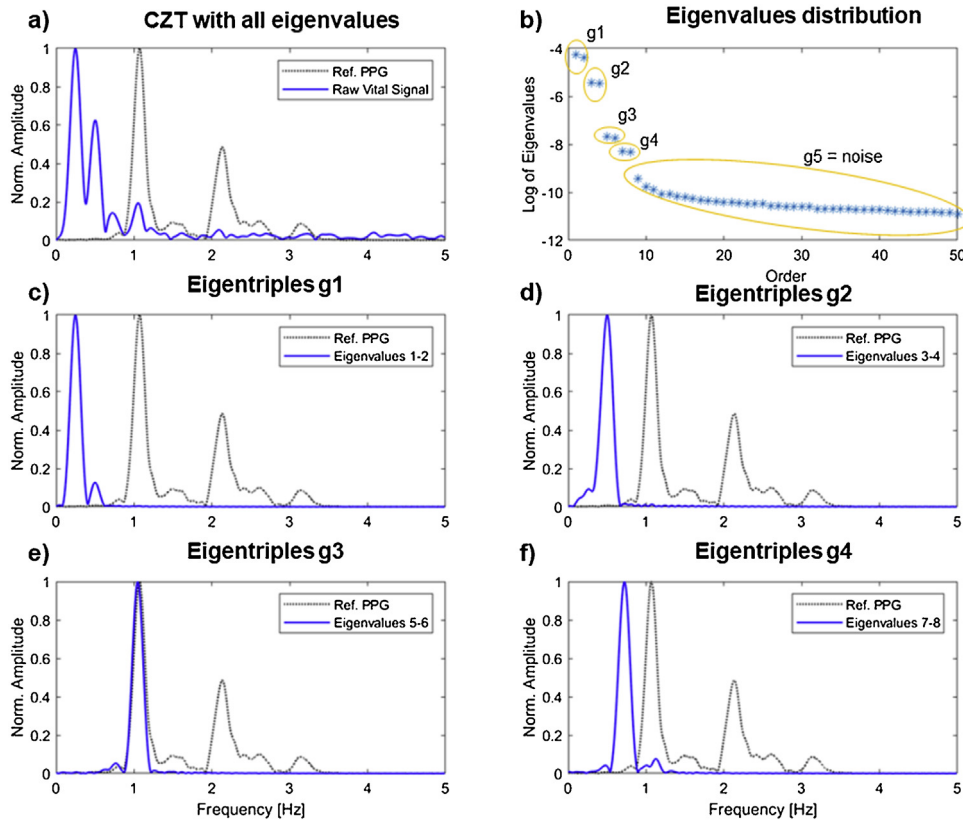


Fig. 4. CZT of vital signal with respect to group of eigenvalues: a) CZT with all eigenvalues, b) distribution of eigenvalues in descending order, c) the first group of eigenvalues g1, d) the second group of eigenvalues g2, e) the third group of eigenvalues g3, and f) the fourth group of eigenvalues g4.

The person was stationary sitting and breathing normally at a fixed distance from the radar. The distances were 0.5, 1.0, 2.0, and 3.0 m. And, different orientations of the person such as front side, 45 degrees, lateral side, and backside to the radar also experimented. A photoplethysmography (PPG, ubpulse 360) [25] sensor was used to measure the heartbeat signal by attaching to a finger for the reference. The UWB radar has a center frequency of 4.6 GHz, bandwidth of 550 MHz [26,27]. The total output power is lower than -41.3 dBm as the Federal Communications Commission (FCC) regulations for consumer electronics in the band of 2.3 GHz at the center frequency of 6.8 GHz. The angle beam of transmitter and receiver antennas is about 75° – 80° , and the data transmission to

PC at a frame rate of $f_s = 44.5$ frames/s. The detail information of the UWB radar is shown in Table 1.

3.2. Vital signs extraction by direct RF sampling

A block of the UWB radar signal was stacked in a matrix $\mathbf{X}$ of 200×512 samples respect to $2\text{m} \times 11.4\text{s}$. The raw signal is presented in Fig. 2(a), which shows the high intensity of stationary clutter signals from furniture, stationary parts of the human body, and antennas coupling. The stationary clutter signal can be easily removed by mean subtraction. Fig. 2(b) presents the mean subtraction (Eq. (12)) data $\bar{\mathbf{X}}$ which shows clearly vital signals. As the

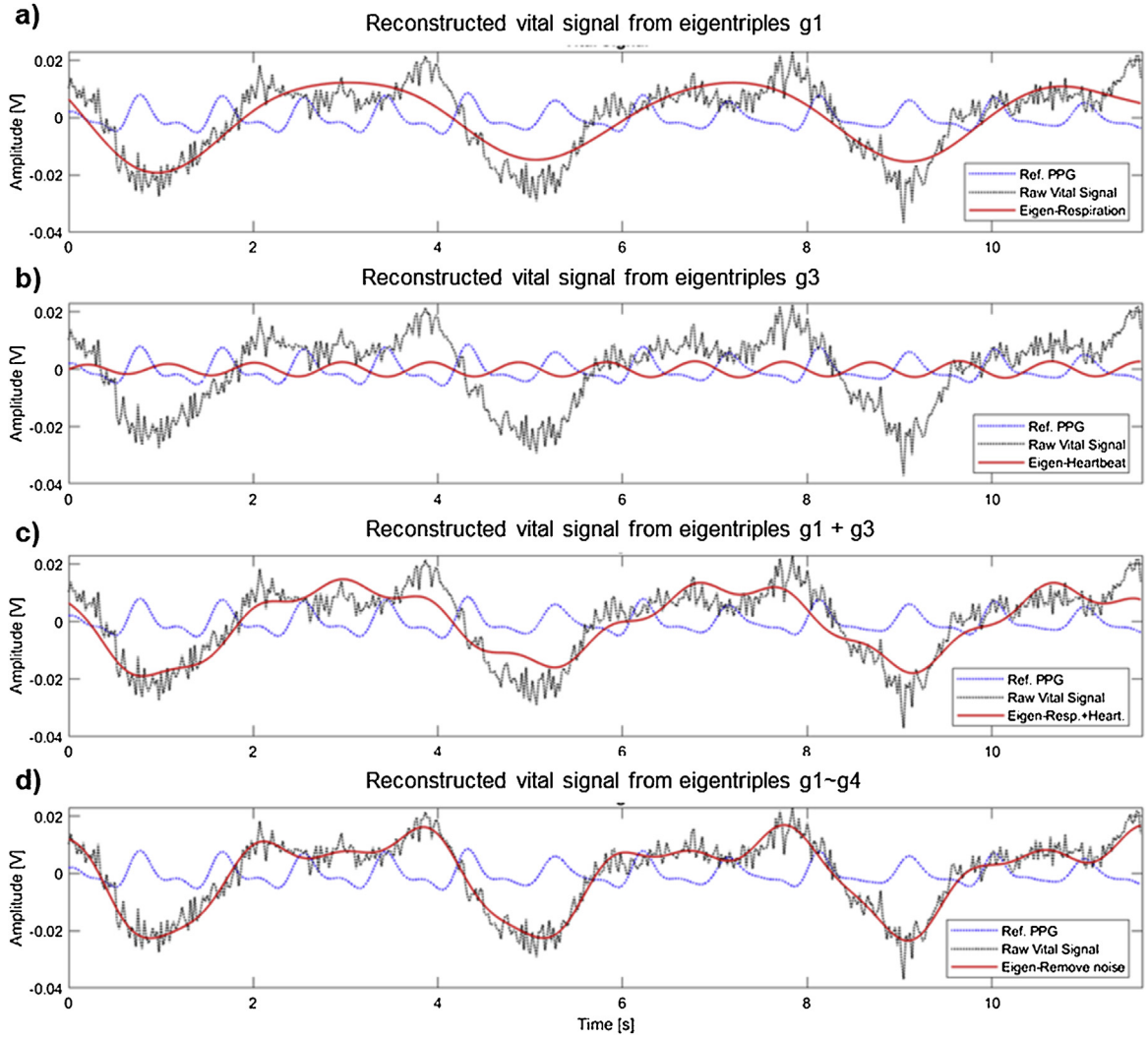


Fig. 5. Reconstructed vital signal from eigentriples: a) eigentriples g1 for the fundamental frequency of respiration signal, b) eigentriples g3 for heartbeat signal, c) combined eigentriples g1 and g3, and d) combined eigentriples g1–g4 for noise removal.

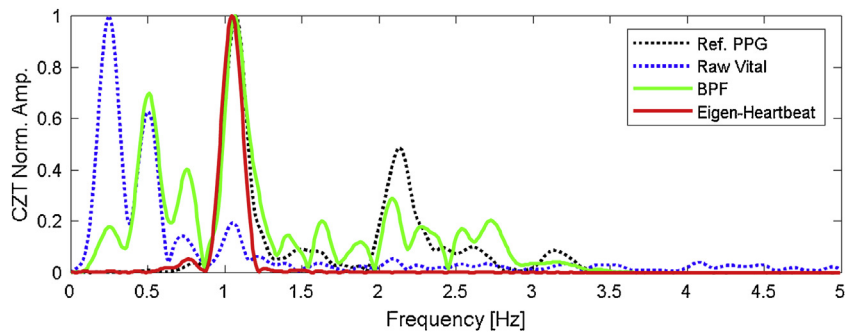


Fig. 6. CZT of heartbeat signal in comparison of the proposed eigenvalue based method with an FIR band-pass filter.

analysis in Section 2, the vital signal is the best quality at the person location, which can be determined by using standard deviation σ in the slow-time direction. The standard deviation σ is expressed in Eq. (13), where $i = 1, 2, \dots, N$ (200) and $j = 1, 2, \dots, M$ (512).

$$\hat{\mathbf{X}}(i, j) = \mathbf{X}(i, j) - \frac{1}{M} \sum_{j=1}^M \mathbf{X}(i, j) \quad (12)$$

$$\sigma(i) = \sqrt{\frac{1}{M} \sum_{j=1}^M \hat{\mathbf{X}}^2(i, j)} \quad (13)$$

For better finding, the person location, the moving average of the standard deviation is used, and the location at the maximum value indicates the person location, as shown in Fig. 3(a). The vital signal x is then extracted from the stacked matrix $\mathbf{X}$ at the person location, as shown in Fig. 2(c). However, it is difficult to see the heartbeat signal due to the high intensity of the respiration signal. Thus, the time-domain signals were transformed into the frequency-domain signal by CZT, as shown in Fig. 3(b). In the frequency domain, the respiration frequency ($f_r = 0.244$ Hz, 14.64 BPM) and its second ($2f_r$)

Table 2
Measurement results of the vital signal at the front side – 1.0 m distance.

Parameters	Raw signal	BPF	Eigentriples g1	Eigentriples g3	Eigentriples g1 + g3	Eigentriples g1–g4	Ref. PPG
SNR of respiration	27.50 dB	25.53 dB	39.70 dB	–	33.23 dB	33.23 dB	–
SNR of heart rate	16.04 dB	16.29 dB	–	34.18	30.84 dB	30.84 dB	21.28 dB

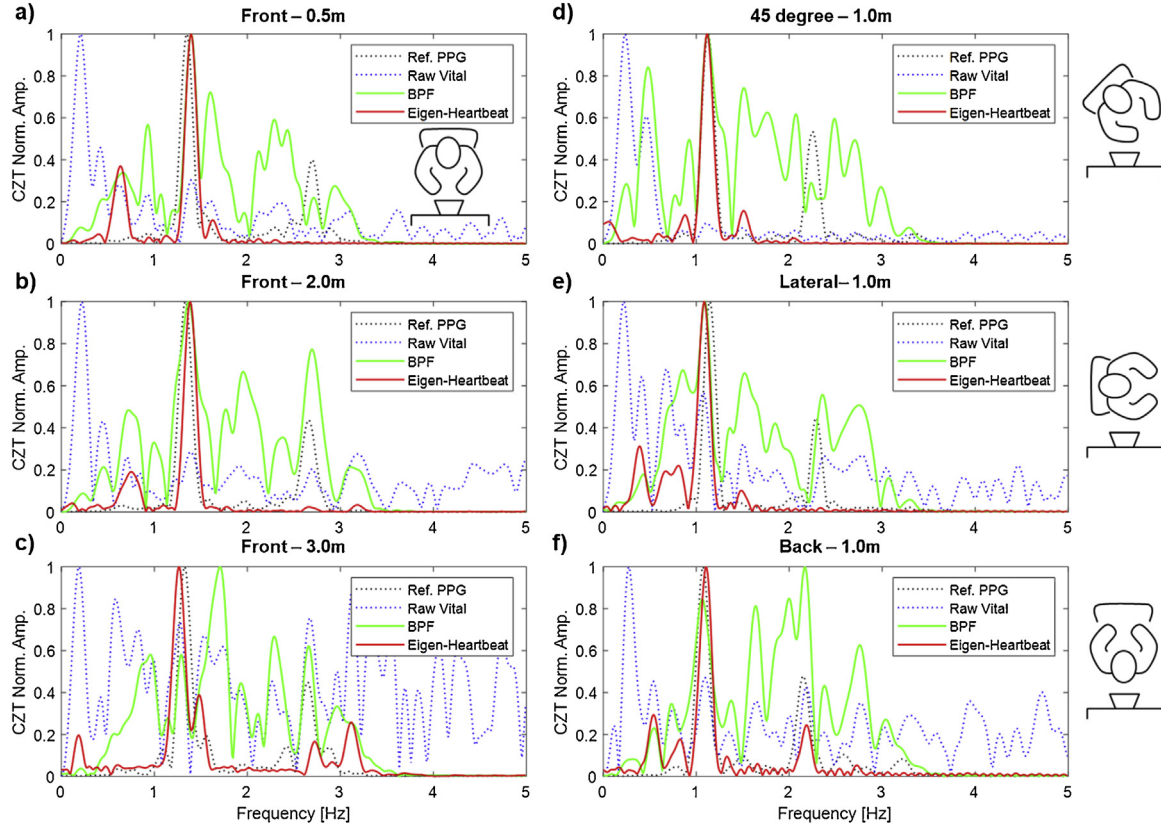


Fig. 7. CZT of heartbeat signal at different distances and orientations of a person: a) front side – 0.5 m, b) front side – 2.0 m, c) front side – 3.0 m, d) 45 degree – 1.0 m, e) lateral side – 1.0 m, and f) backside – 1.0 m.

and third ($3f_r$) harmonic frequencies can be seen as pointed in Fig. 3(b). Heart rate (f_h) also can be seen at the peak of 1.0449 Hz (62.69 BPM), similar to 1.0723 Hz (64.34 BPM) of the PPG measurement. In the next section, extraction of the fundamental and harmonics of respiration frequencies, and heartbeat frequency will be better found in relation with the eigenvalues.

3.3. Vital signs extraction by eigenvalues

Taking vital signal x in Fig. 2(c) into the analysis of the eigenvalue based method in Section 2.3. The purpose is extraction of vital signal components such as fundamental and harmonics of respiration signal and heartbeat signal by analysis of eigenvalue distribution. Trajectory matrix H is first formed in a size of 512×513 matrix. Then, the covariance matrix C is computed as Eq. (8). The eigenvalues λ_i are sorted in descending order. The first 50 eigenvalues λ_i are shown in Fig. 4(b). It is observed that the eigenvalues are grouped into 5 groups, which are related to the peaks in the CZT graph of Fig. 4(a). Each group has two eigenvalues with a similar value that represents components of vital signal, and the last group includes the small eigenvalues for the noise signal. The first group (g1) shows the fundamental frequency of respiration, as shown in Fig. 4(c). The second group (g2) shows the second harmonic frequency of respiration, as shown in Fig. 4(d). The third group (g3) shows the heartbeat frequency, as shown in Fig. 4(e). The fourth

group (g4) shows the third harmonic frequency of respiration, as shown in Fig. 4(f). And, the fifth group (g5) represents for noise signal which is not shown here. So, by the selection of eigenvalues (λ_i) with respect eigenvectors (U_i, V_i) to build eigentriples group and reconstruction back to the original signal by Eq. (11), the fundamentals and harmonics of the respiration signal and heartbeat signal could be retrieved.

The reconstructed vital signals by the selection of eigentriples are shown in Fig. 5. The reconstruction is calculated by Eq. (11). Reconstruction by the first eigentriples group (g1), which represents the fundamental frequency of the respiration signal, is shown in Fig. 5(a). It shows a high quality of respiration signal compared to the noisy raw signal. The heartbeat signal was reconstructed by the third eigentriples group (g3), as shown in Fig. 5(b). A good agreement of the reconstructed heartbeat signal with reference PPG sensor can be observed. The combination of the respiration signal and heartbeat signal is shown in Fig. 5(c), which is the reconstructed signal from eigentriples g1 and g3. Fig. 5(d) shows the total vital signal excluding the noise signal of eigentriples g5, which was reconstructed by eigentriples g1–g4. The noise removal process can be done by setting a threshold of eigenvalues (Fig. 4(b)) to separate the signal and noise. The eigenvalues of noise signal have small values and smoothly decrease as their order increase.

To show the advantages of the proposed eigenvalue based method, a high-order FIR band-pass filter (BPF) was used. The BPF

Table 3

Signal to noise ratio of heartbeat signal at different distances and orientations.

Orientation-distance	Raw signal	BPF method	Proposed eigenvalue based method	Ref. PPG
Front – 0.5 m	9.01 dB	9.70 dB	32.28 dB	24.35 dB
Front – 1.0 m	16.02 dB	17.06 dB	35.59 dB	21.59 dB
Front – 2.0 m	7.79 dB	9.09 dB	30.54 dB	26.41 dB
Front – 3.0 m	4.83 dB	2.98 dB	21.75 dB	21.87 dB
45 deg – 1.0 m	7.45 dB	7.54 dB	29.40 dB	27.32 dB
Lateral – 1.0 m	8.92 dB	7.42 dB	27.94 dB	28.16 dB
Back – 1.0 m	7.95 dB	6.29 dB	26.81 dB	24.23 dB

has an order of 128 and bandwidth of 0.1–0.7 Hz and 0.75–3.0 Hz, which are the normal range of respiration and heart rate of a healthy person, respectively. Fig. 6 shows the CZT of different signals including reference PPG signal, the raw signal of UWB radar, BPF signal, and proposed eigenvalue based method. It is easy to observe the high quality of the signal processed by the proposed eigenvalue based method signal compared to the BPF signal and raw signal. The quantitative signal-to-noise (SNR) is taken into account, as expressed in Eq. (14) [28], where $CZT_{resp/hr}$ is the amplitude of CZT transform of the fundamental respiration/heart rate; CZT_{noise} of respiration/heartbeat is the amplitude of CZT transform of the noise signal, which is all the frequencies in 0.1–0.7 Hz/0.75 Hz–3.0 Hz excluding fundamental and harmonics of respiration/heart rate. The SNR of respiration/heart rate was improved about 12 dB/18 dB using the eigentriples g1/g3 compared to the raw signal and BPF methods. The SNR of the combined eigentriples g1 and g3 and eigentriples g1–g4 shows about 8 dB/14 dB better than the raw signal and BPF signal. This was reduced about 4 dB compared to the single group because of the interference of respiration and heartbeat signals. And, the measurement of heart rate is similar for all the approaches (1.0449 Hz, 62.69 BPM) which are about 2.56% error compared to the result of the reference PPG sensor (1.0723 Hz, 64.34 BPM). The detail measurement results are shown in Table 2.

$$SNR = 20 \cdot \log \frac{CZT_{resp/hr}}{\sqrt{\frac{1}{N} \sum_{i=1}^N CZT_{noise-i}^2}} \quad (14)$$

For further illustrating the performance of the proposed eigenvalue based method in more noisy situations, experiments on different orientations of a person (sitting 45 degrees, lateral side, and backside) and different distances of the person to the UWB radar (0.5 m–3.0 m) were done and shown in Fig. 7. The results show that the proposed eigenvalue based method provides much better SNR compare to the BPF method. The detail SNR is shown in Table 3. The SNR of the raw and BPF vital signal decreases as the distance increases or less thorax volume coverage the radar beam (in order: backside, lateral side, 45 degrees, to front side). However, the SNR was much improved using the proposed eigenvalue based method. The influence of the harmonic components of the respiration signal could be observed at different distances of the person to the UWB radar. At near distance, the heartbeat signal was influenced by the second harmonic respiration signal which is the first peak signal (around 0.7 Hz) in the proposed eigenvalue based method signal of Fig. 7(a). As the distance increase (2.0 m and 3.0 m), the influence of the respiration signal on the heartbeat signal was reduced; but, the SNR of both respiration and heartbeat signal decreases.

4. Conclusion & discussion

This paper presents a frequency extraction method based on eigenvalues for remote vital measurement of UWB radar. The vital signal was arranged in the form of Hankel's matrix, and

the eigentriples, including eigenvalues and eigenvectors, were decomposed. The frequencies of the vital signal, which are fundamental/harmonic frequencies of respiration and frequency of heartbeat, could be exactly extracted by the groups of eigentriples. Each group of eigentriple has two eigenvalues that have a similar value. The proposed method could help to improve the SNR of 12 dB and 18 dB for respiration and heartbeat compared to the conventional FIR filters, respectively, for different thorax orientations and distances from 0.5 m to 3.0 m. In addition, the noise signal also could be grouped and removed easily by excluding the eigenvalues below a fixed threshold. The approach is potential for use in monitoring of vital signals while a person is relaxing or sleeping without movement; in those cases, a movement detector should be implemented to detect the movement state of the person to pause the vital measurement. In future work, an auto-grouping and selecting eigentriple groups for respiration and heartbeat frequencies should be carried out. This will allow the proposed algorithm operates automatically.

Declaration of Competing Interest

There is no conflict of interest.

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Biography



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